Chapter:5
DC Machine

notes.arpanadhikari7.com.np

Available on notes.arpanadhikari7.com.np

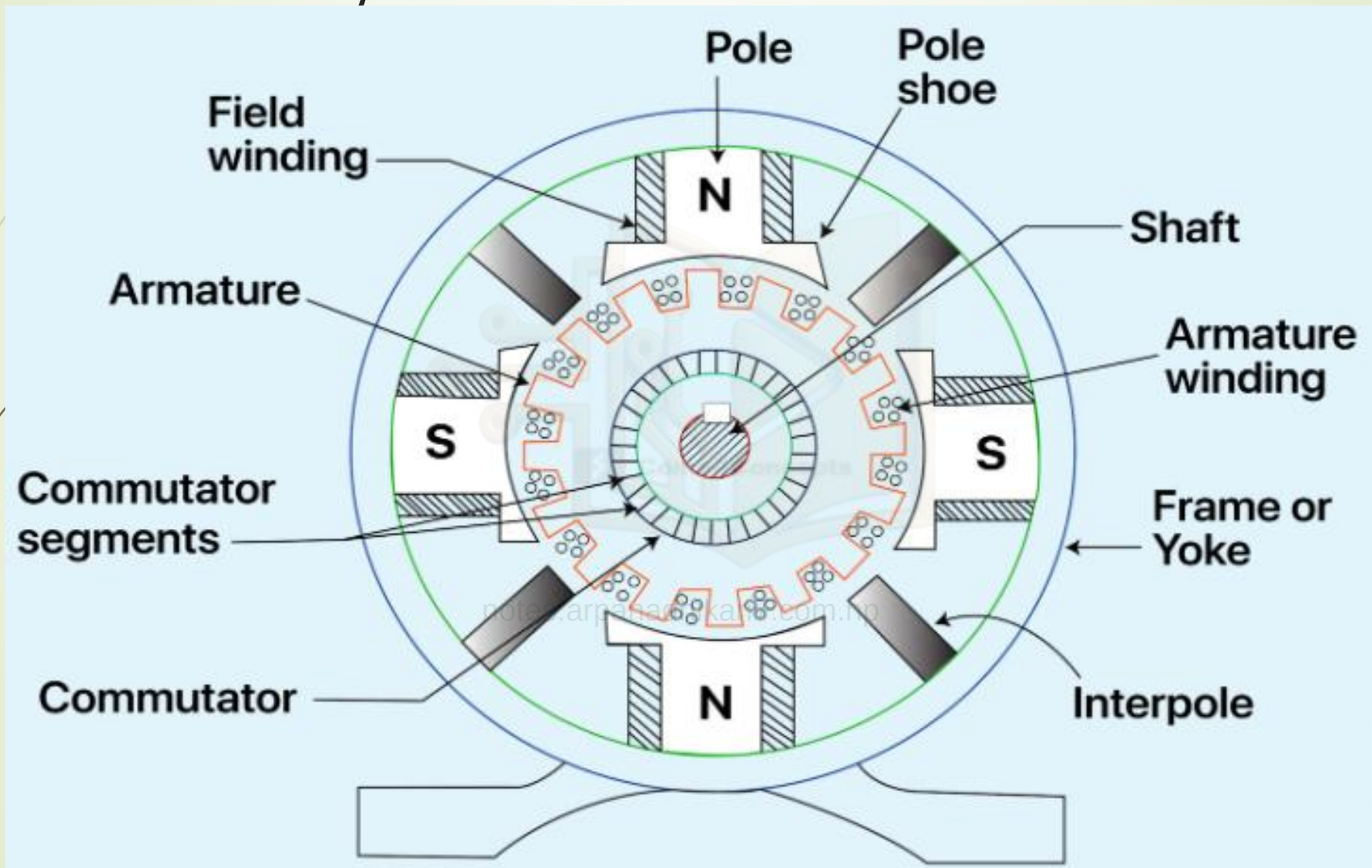
Introduction

- DC machine is an **electromechanical energy alteration device**
- DC machine that converts mechanical power to electrical power are Generators, where as a DC machine that converts electrical power to mechanical power are Motors
- DC machine is most frequently used for a motor
- The main benefits of this machine include torque regulation as well as easy speed regulation.
- The **applications of the DC machine** is limited to trains, mills, and mines
- For example, underground subway cars, as well as trolleys, may utilize DC motors. In the past, automobiles were designed with DC dynamos for charging their batteries

notes.arpanadhikari7.com.np

Available on notes.arpanadhikari7.com.np

Construction of DC machines(Generator and Motor)



Available on notes.arpanadhikari7.com.np

1. Yoke (Magnetic Frame)

- The outer frame or yoke serves a double purpose
- It provides **mechanical support** for the poles and acts as a protecting cover for the whole machine
- It **carries the magnetic flux** produced by the poles
- yokes are made of **cast iron** or **steel**

2. Field system(Stator)

- Stator is the **stationary part** of the machine
- Its main function is to **provide main magnetic flux** in which **armature coil rotates**.
- It consists of poles and pole shoes

Pole core and Pole Shoes

- The field magnets consist of **poles (north south poles of magnet) and pole shoes**.
- The pole shoes serve two purposes:

I. they **spread out the flux** in the air gap

II. they support the **exciting coils (or field coils)**

notes.arpanadhikari7.com.np

Available on notes.arpanadhikari7.com.np

Field windings

- It is the winding that produces **main flux** in the machine.
- Field winding is applied to **stator**.
- These windings are **wound on poles** in such a way that whenever current passes through it they act like **electromagnet**.

2. Armature System(Rotor)

- Rotor is the **basic part** of DC machine.
- It consists of armature core, armature winding and commutator.

Armature

- It is the **rotating part** of dc machine.
- It is **cylindrical** in shape.
- It consists of **core that is laminated** to reduce eddy current loss.
- **Core slots** are also present that **holds armature windings**

notes.arpanadhikari7.com.np

Available on notes.arpanadhikari7.com.np

Armature windings

- These are the conductor **winding placed in the slots** provided in the armature core.
- When armature core rotates, **EMF is induced** in the armature winding.
- **Coil ends** of armature windings are connected to **commutator segments.**

3. Commutator

- It is a **mechanical rectifier** used for obtaining a rectified waveform at the output in order to get dc voltage.
- It is responsible to **supply current** to the **load** through brushes.
- The commutator is made of **copper segments** insulated from each other by mica sheets and mounted on the shaft of the machine.

notes.arpanadhikari7.com.np

4. Brushes

- It helps to **gather the current from the commutator and supply it to the exterior load.**
- **It acts as a contact.**

Available on notes.arpanadhikari7.com.np

DC Generator

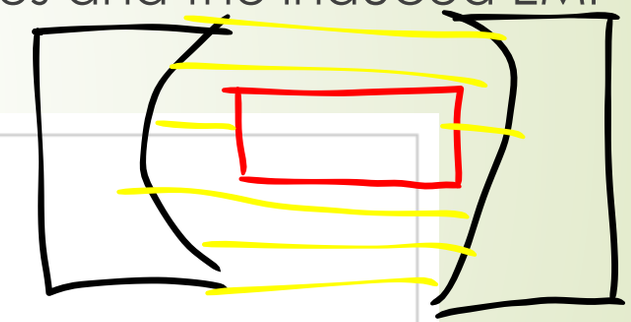
- It is an **electromechanical energy** conversion device that converts **mechanical power into DC electrical power** through the process of electromagnetic induction.
- A DC generator operates on the **principle of electromagnetic induction**
- When the magnetic flux linking a conductor changes, an EMF is induced in the conductor.
- A DC generator has a **field winding and an armature winding**.
- The **armature winding** of DC Generator is placed on the **rotor** whereas the **field winding** is placed on the **stator**.
- **EMF induced** in the armature winding of a **DC generator is alternating** and is converted into **direct voltage using a commutator**

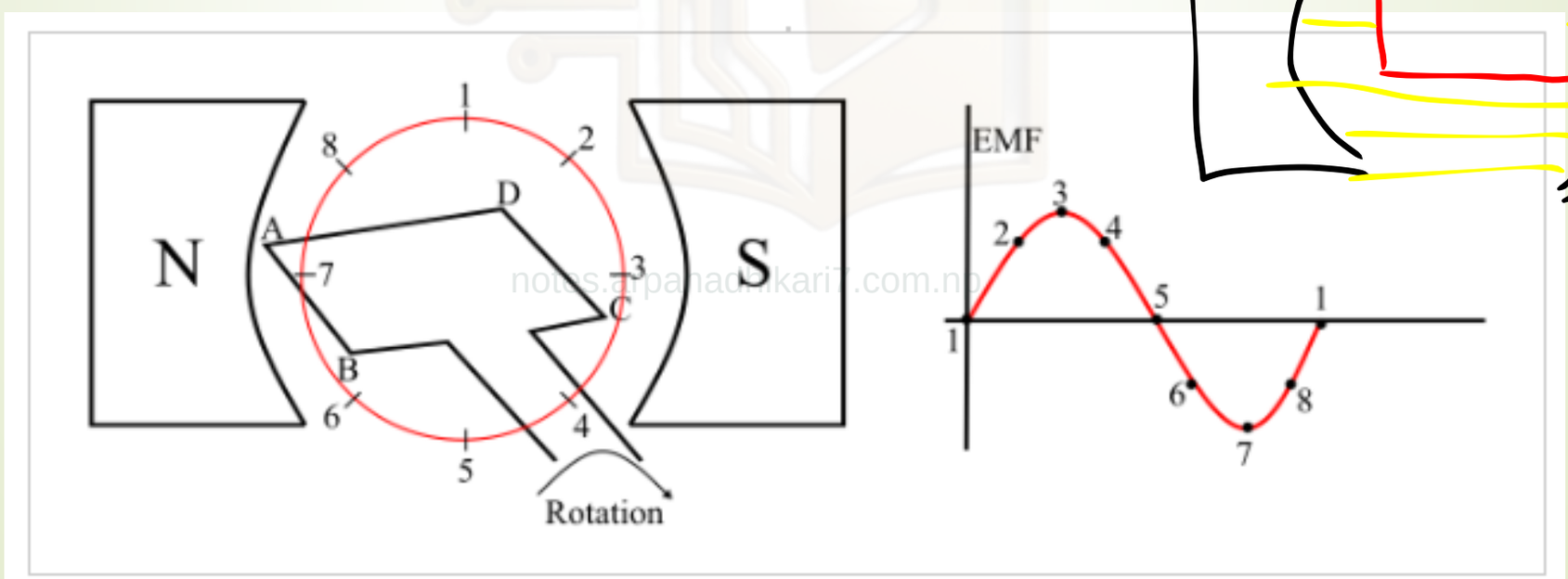
notes.arpanadhikari7.com.np

Available on notes.arpanadhikari7.com.np

Working principle of DC generator

- Consider a single loop DC generator (as shown in the figure), in this a single turn loop 'ABCD' is **rotating clockwise** in a uniform magnetic field with a constant speed.
- When the loop rotates, the magnetic flux linking the coil sides 'AB' and 'CD' changes continuously.
- This **change in flux linkage** induces an **EMF in coil** sides and the induced EMF in one coil side adds the induced EMF in the other.





notes.arpanadhikari7.com.np

Available on notes.arpanadhikari7.com.np

The EMF induced in a DC generator can be explained as follows

- When the loop is in position-1, **the generated EMF is zero** because, the movement of coil sides is parallel to the magnetic flux.
- When the loop is in position-2, the coil sides are moving at an angle to the magnetic flux and hence, a **small EMF is generated.**
- When the loop is in position-3, the coil sides are moving at **right angle** to the magnetic flux, therefore the **generated EMF is maximum**.
- When the loop is in position-4, the coil sides are cutting the magnetic flux at an angle, thus a **reduced EMF** is generated in the coil sides.
- When the loop is in position-5, no flux linkage with the coil side and are **moving parallel to the magnetic flux**. Therefore, no EMF is generated in the coil.
- At the position-6, the coil sides move under a pole of opposite polarity and hence the **polarity of generated EMF is reversed**. The maximum EMF will generate in this direction at position-7 and zero when at position-1.
- This cycle repeats with revolution of the coil.

notes.arpanadhikari7.com.np

Available on notes.arpanadhikari7.com.np

- It is clear that the generated EMF in the loop is **alternating one**.
- It is because any coil side (say AB) has EMF in one direction when under the influence of N-pole and in the other direction when under the influence of S-pole.
- Hence, when **a load** is connected across the terminals of the generator, **an alternating current will flow through it.**
- And by using a commutator, this **alternating emf generated in the loop** can be converted into direct voltage

notes.arpanadhikari7.com.np

Available on notes.arpanadhikari7.com.np

EMF Equation

- Let us suppose there are Z total numbers of conductor in a generator, and arranged in such a manner that all parallel paths are always in series.
  Here,
  Z = total numbers of conductor
  A = number of parallel paths
  Then,
  Z/A = number of conductors connected in series
  We know that induced emf in each path is same across the line
  Therefore,
  Induced emf of DC generator
  e= emf of one conductor × number of conductor connected in series.
- A=P for lap type of winding, A= 2 for wave type winding, p= number of poles
- Induced emf of DC generator is,

$$e = \phi P \frac{N}{60} X \frac{Z}{A} volts$$

notes.arpanadhikari7.com.np

Available on notes.arpanadhikari7.com.np

Types of DC Generator

- DC generators are classified based on how their fields are excited (i.e. produced). There are three methods of excitation, and thus three main types of DC generators:
- Permanent Magnet DC Generators :**Field coils excited by permanent magnets**
- Separately Excited DC Generators : **Field coils excited by some external source**
- Self Excited DC Generators: **Field coils excited by the generator itself**

Self-excited DC generators can further be classified depending on the position of their field coils. The three types of self-excited DC generators are:

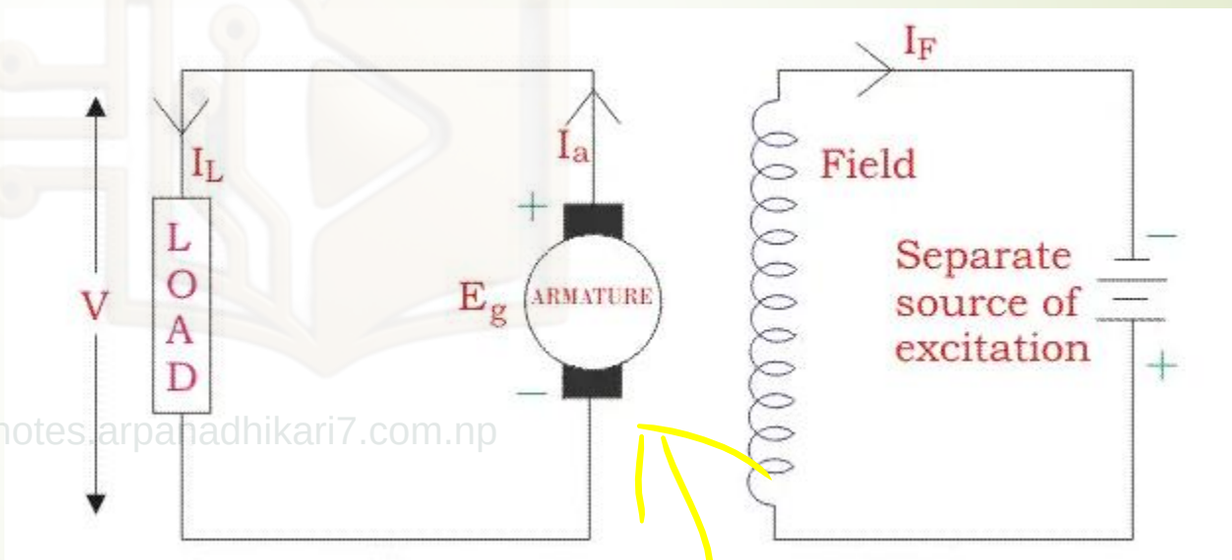
- Series Wound Generators
- Shunt Wound Generators
- Compound Wound Generators

notes.arpanadhikari7.com.np

Available on notes.arpanadhikari7.com.np

Separately Excited DC Generator

- These are the generators whose field magnets are energized by some external DC source, such as a battery.
- A circuit diagram of separately excited DC generator is shown in the figure below. The symbols below are:
- I_a = Armature current
- I_L = Load current
- V = Terminal voltage
- E_g = Generated EMF



notes.arpanadhikari7.com.np

$E_g = I_a R_a + V$

R_a

Available on notes.arpanadhikari7.com.np

- Voltage drop in armature = Ia*Ra
- Let, $I_a = I_L = I\ (say)$
- Then, ~~voltage across the load, $V = IR_a$~~

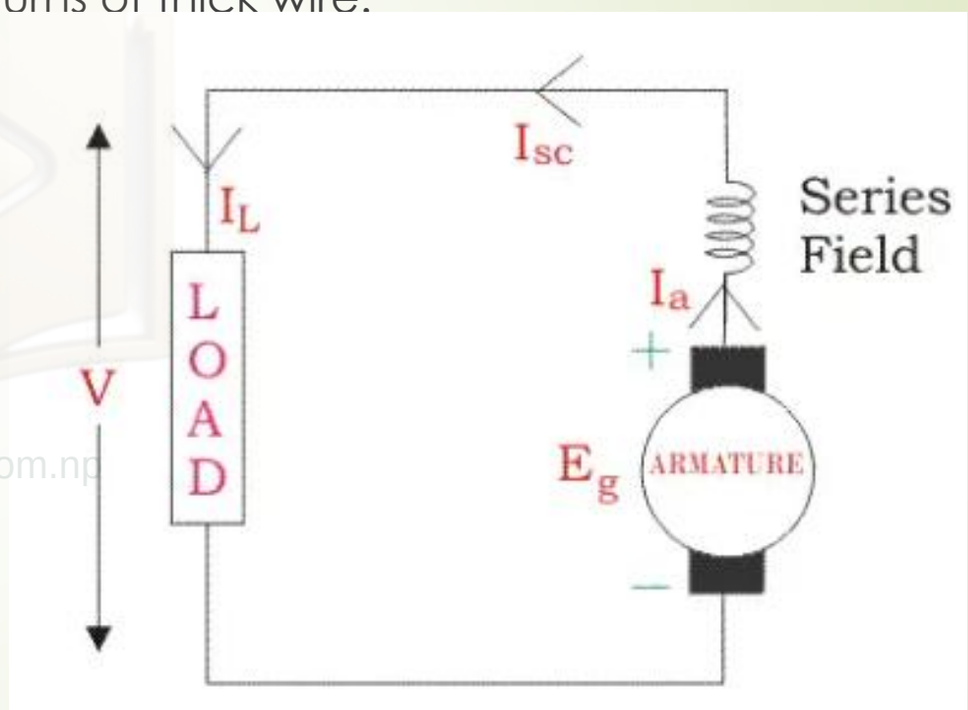
- Power Generated is equal to, $P_g = E_g \times I$
- Power delivered to external load is equal to, $P_L = V \times I$

notes.arpanadhikari7.com.np

Available on notes.arpanadhikari7.com.np

Self Excited DC Generators

- **Series Wound Generator**
- In these type of generators, **the field windings are connected in series with armature** conductors, as shown in the figure below.
- Whole current flows through the field coils as well as the load. As series field winding carries full load current it is designed with relatively few turns of thick wire.
- Here:
- R_{sc} = Series winding resistance
- I_{sc} = Current flowing through the series field
- R_a = Armature resistance
- I_a = Armature current
- I_L = Load current
- V = Terminal voltage
- E_g = Generated EMF



notes.arpanadhikari7.com.np

Available on notes.arpanadhikari7.com.np

- Then,

$$I_a = I_{sc} = I_L = I\ (say)$$

- Voltage across the Load is equal to,

$$V = E_G - I(R_a + R_{se})$$

- Power generated is equal to ,

$$P_g = E_g \times I$$

- Power delivered to load,

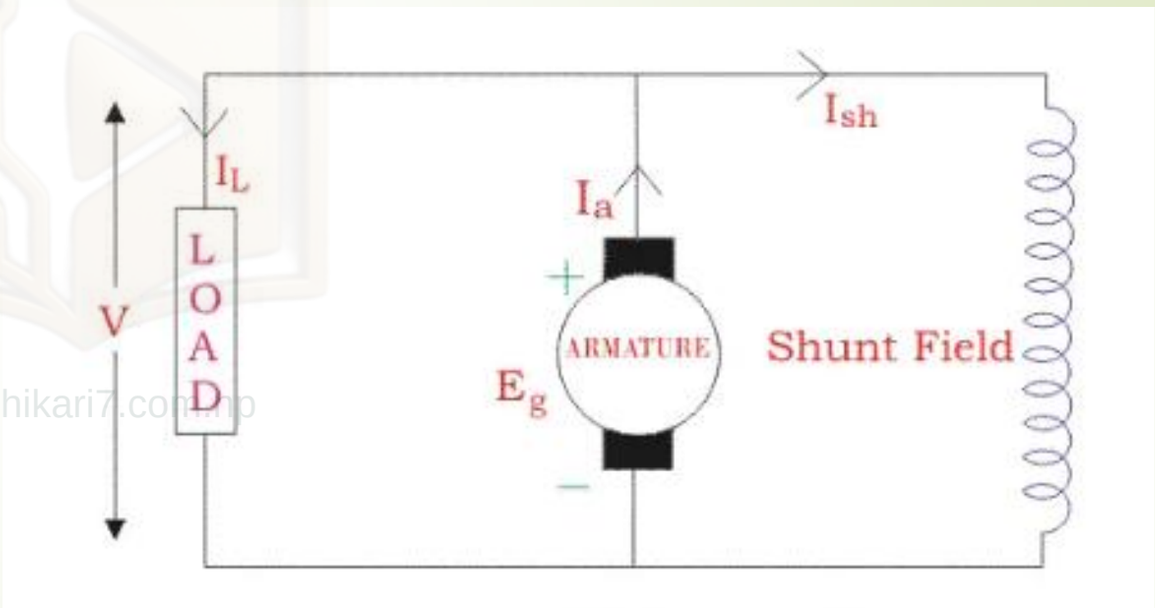
$$P_L = V \times I$$

notes.arpanadhikari7.com.np

$$E_g = I_a(R_a + R_s) + V$$

Available on notes.arpanadhikari7.com.np

- **Shunt Wound DC Generators**
- In these type of DC generators, the **field windings are connected in parallel with armature conductors**, as shown in the figure below. In shunt wound generators the voltage in the field winding is same as the voltage across the terminal.
- Here:
- R_{sh} = Shunt winding resistance
- I_{sh} = Current flowing through the shunt field
- R_a = Armature resistance
- I_a = Armature current
- I_L = Load current
- V = Terminal voltage
- E_g = Generated EMF



notes.arpanadhikari7.com.np

Available on notes.arpanadhikari7.com.np

- Here armature current I_a is dividing in two parts – one is shunt field current I_{sh}, and another is load current I_L.
- So,

$$I_a = I_{sh} + I_L$$

- Shunt field current,

$$I_{sh} = \frac{V}{R_{sh}}$$

- Voltage across the load

$$V = E_g - I_a R_a$$

- Power generated is

$$P_g = E_g \times I_a$$

- Power delivered to the load is,

$$P_L = V \times I_L$$

notes.arpanadhikari7.com.np

$I_a = I_{sh} + I_L$

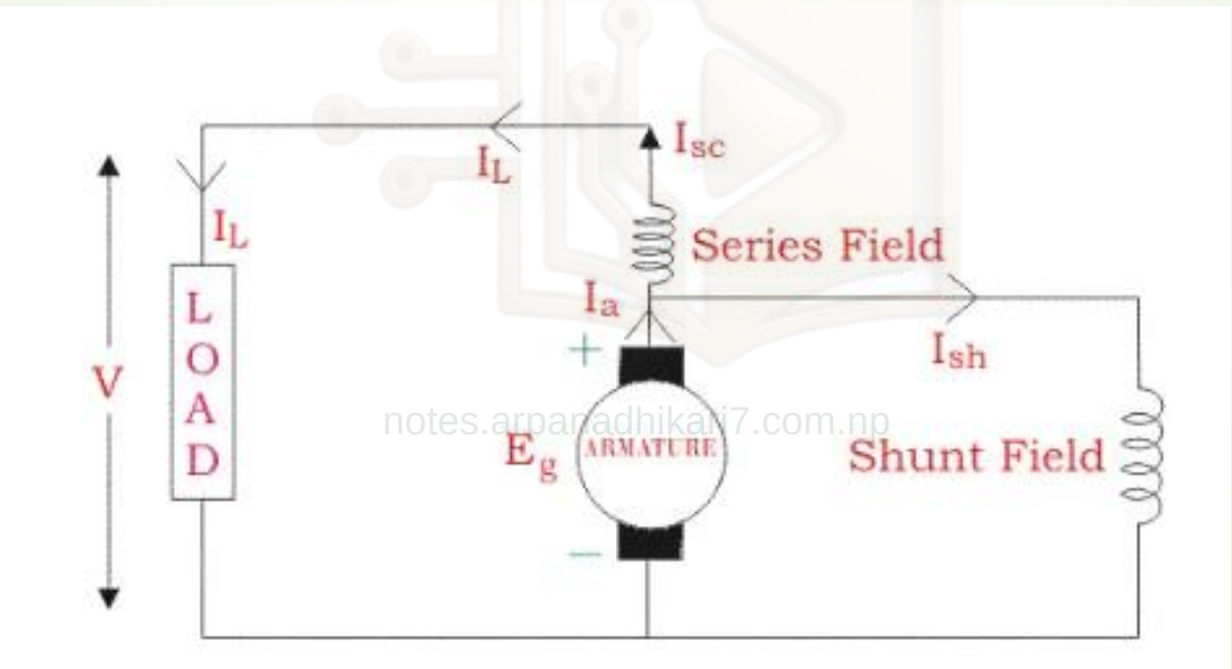
$E_g - I_a R_a - V = 0$

Available on notes.arpanadhikari7.com.np

- **Compound Wound DC Generator**
- This type of DC generators may be of two types- short shunt compound-wound generator and long shunt compound-wound generator.

Short Shunt Compound Wound DC Generator

- Short Shunt Compound Wound DC Generators are generators **where only the shunt field winding is in parallel with the armature winding**, as shown in the figure below.



notes.arpanadhikari7.com.np

Available on notes.arpanadhikari7.com.np

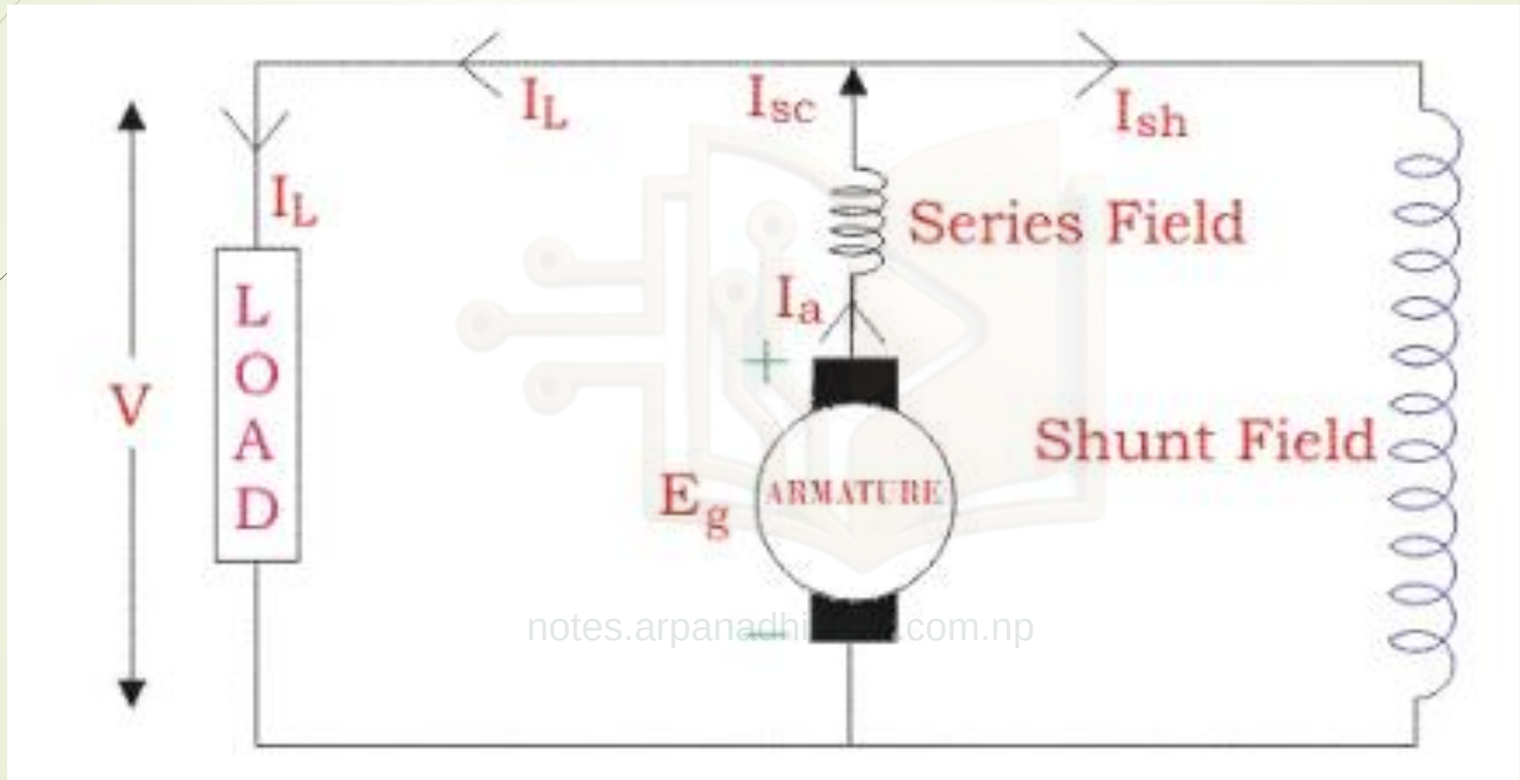
- Series Field current is, $I_{sc} = I_L$
- Shunt field current is, $I_{sh} = \frac{(V + I_{sc}R_{sc})}{R_{sh}}$
- Armature current is, $I_a = I_{sh} + I_L$
- Voltage across the load is, $V = E_g - I_aR_a - I_{sc}R_{sc}$
- Power generated is, $P_g = E_g \times I_a$
- Power delivered to load is, $P_L = V \times I_L$

notes.arpanadhikari7.com.np

Available on notes.arpanadhikari7.com.np

Long Shunt Compound Wound DC Generator

- Long Shunt Compound Wound DC Generator are generators where the **shunt field winding is in parallel with both series field and armature winding**, as shown in the figure below.



Available on notes.arpanadhikari7.com.np

- Shunt field current, $$I_{sh} = \frac{V}{R_{sh}}$$
- Armature current, $$I_{sc} = I_L + I_{sh}$$
- Voltage across the load is, $$V = E_g - I_a R_a - I_{sc} R_{sc} = E_g - I_a(R_a + R_{sc}) \ [\because I_a = I_{cs}]$$
- Power generated is, $$P_g = E_g \times I_a$$
- Power delivered is, $$P_L = V \times I_L$$

$$E_g - I_a R_a - I_{sh} R_{sh} - V = 0$$

$$I_{sh} = \frac{V}{R_{sh}}$$

notes.arpanadhikari7.com.np

Available on notes.arpanadhikari7.com.np

DC motor and its working principle

- A DC motor is an **electromechanical energy conversion device**, which converts **electrical energy input into the mechanical energy** output.
- Consider a two pole DC motor. When the DC motor is connected to an external source of DC supply, the field coils are excited developing alternate N and S poles and a current flows through the armature windings.
- All the armature conductors under **N pole carry current in one direction** whereas all the conductors under **S pole carry current in the opposite direction.**
- As each **conductor carrying a current is placed in a magnetic field**, hence a **mechanical force acts** on it.
- The **force on all the conductors add together to exert a torque** which make the armature rotating.

notes.arpanadhikari7.com.np

Available on notes.arpanadhikari7.com.np

Motor torque

- Torque is the **tendency of a force to rotate or move** an object about an axis.
- A force is a push or pull, likewise, torque is a twist to an object. Mathematically, torque, T = F × r.
- Let T_g = armature or gross torque (N-m) = Force × radius.
- r = radius of the armature in m.
- N = speed of the armature in rpm = N/60 rps.

Work done/revolution = force × distance moved per revolution

$$w.d. = F \times 2\pi r \ Nm$$

$$w.d./s = F \times 2\pi r \times \frac{N}{60} \ Nm$$

notes.arpanadhikari7.com.np

$$w.d. = \frac{2\pi N}{60}(F \times r) \ Nm/s \ or \ watt$$

$$w.d. = \frac{2\pi N T_g}{60} \ watt \ \ldots.(1)$$

Available on notes.arpanadhikari7.com.np

- The expression for voltage in dc motor is given by,

$$V = E_a + I_a R_a$$

$$(mutiply\ throughout\ by\ I_a)$$

$$V I_a = E_a I_a + I_a^2 R_a$$

- *Electrical input = electrical power equivalent to mechanical power developed + armature copper loss*
- Mechanical power developed,

$$= E_a I_a \ watt \ \ldots.(2)$$

- Since equation (1) = equation (2),

$$\frac{2\pi N T_g}{60} = E_b I_a$$

$$\frac{2\pi N T_g}{60} = \frac{\phi Z N}{60} \times \frac{P}{A} \times I_a$$

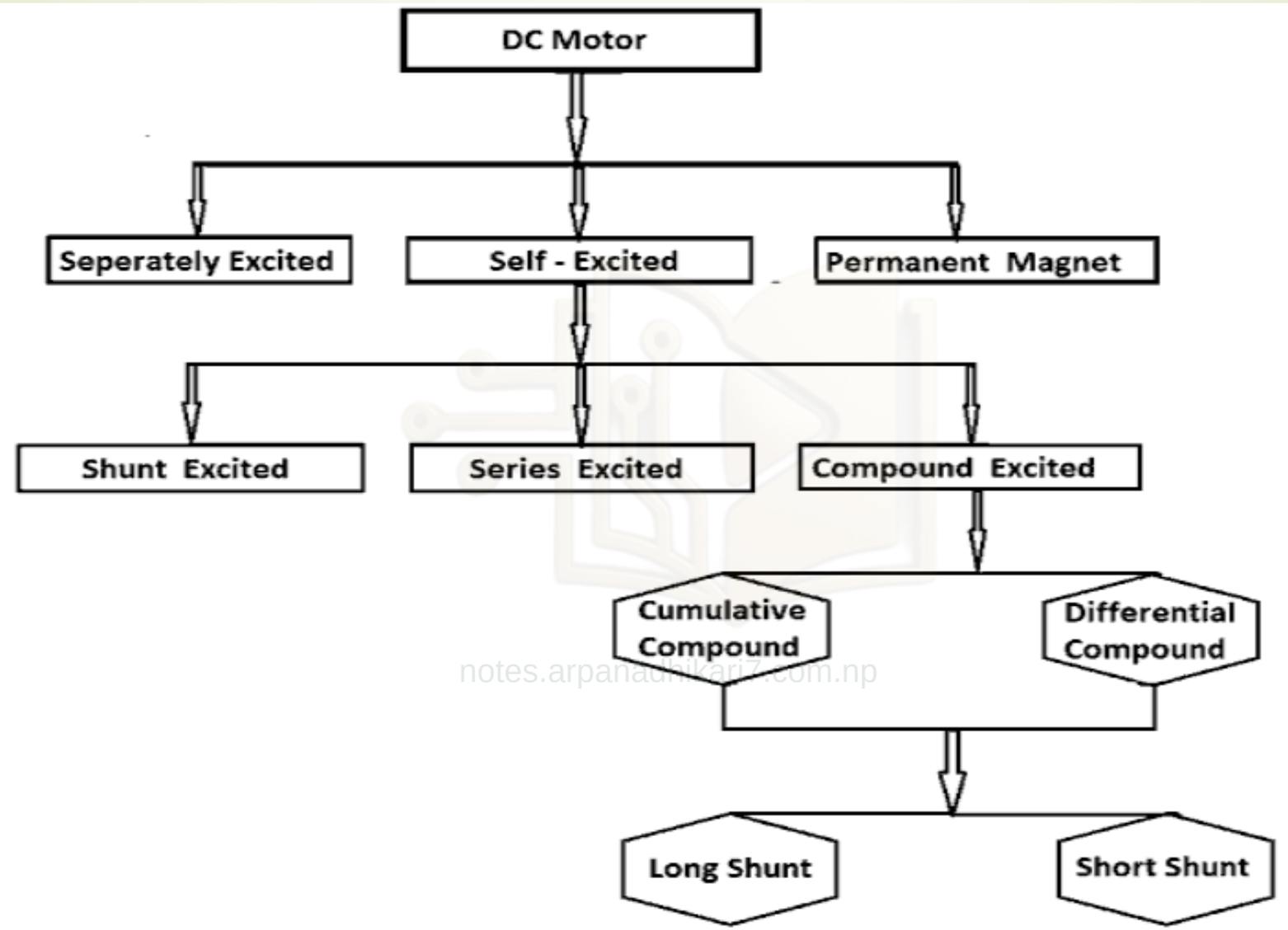
$$T_g = \frac{1}{2\pi} \phi Z I_a \times \frac{P}{A}$$

$$T_g = 0.159\, \phi Z I_a \times \frac{P}{A} \ Nm$$

notes.arpanadhikari7.com.np

Available on notes.arpanadhikari7.com.np

Types of DC motors

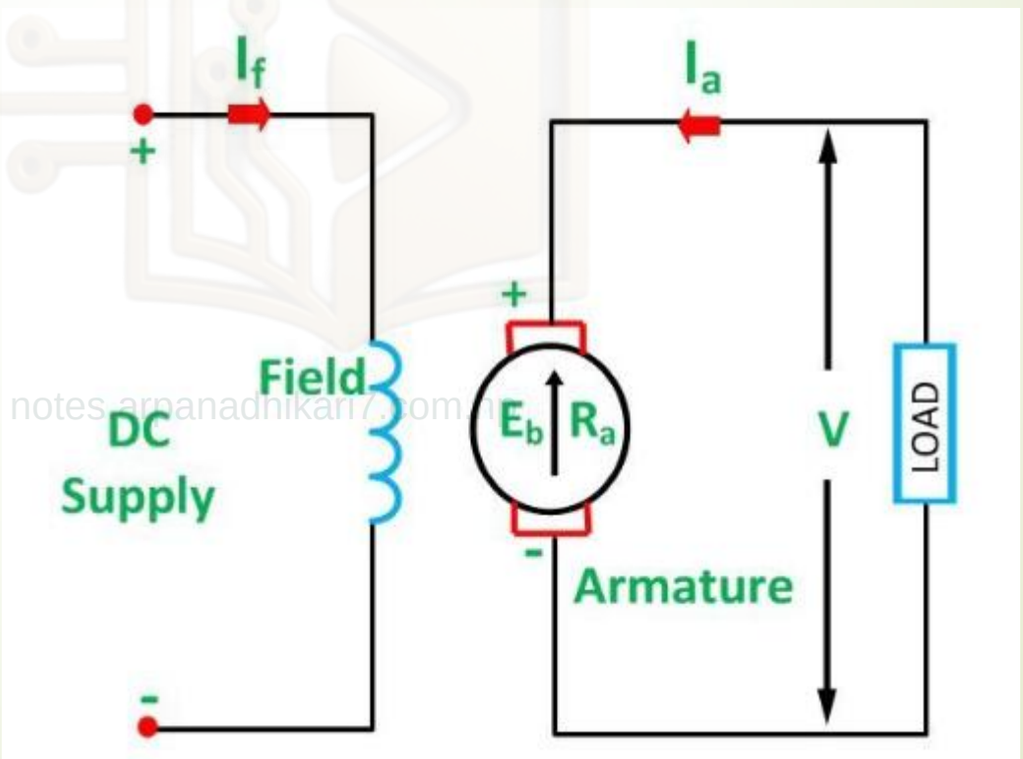


notes.arpanadhikari7.com.np

Available on notes.arpanadhikari7.com.np

Separately Excited DC Motor

- The supply is given separately to the field windings.
- The armature current does not flow through the field windings, as the field winding is energized from a separate external source .
- We know $T_g = K_a \varphi I_a$
- So the torque in this case can be varied by varying field flux φ, independent of the armature current I_a.



notes.arpanadhikari7.com.np

Available on notes.arpanadhikari7.com.np

Self Excited DC Motor

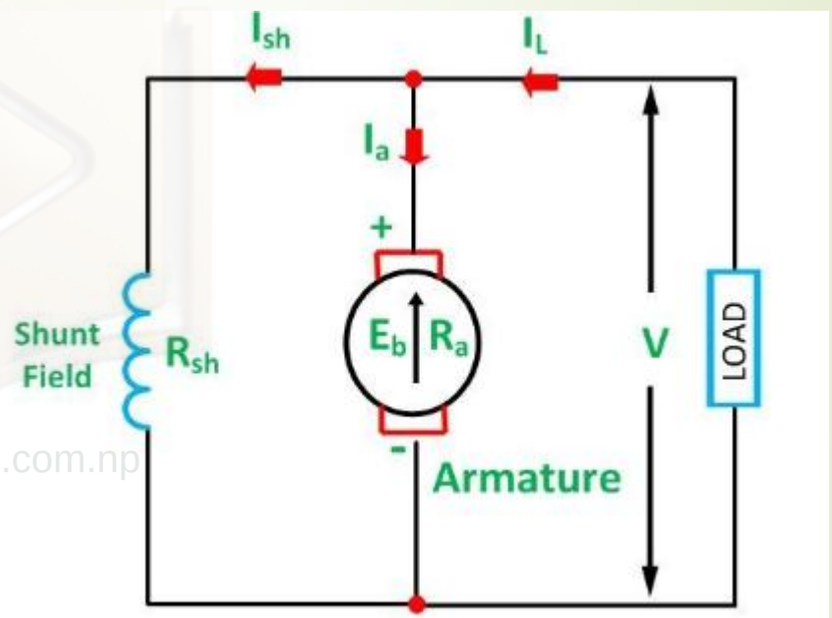
- In this type of motor, the current in the windings is supplied by the machine or motor itself.

1. Shunt Wound Motor

- This is the most common types of DC Motor. Here the field winding is connected in parallel with the armature.
- On applying KCL at the shown junction

$$I = I_a + I_{sh} \ldots\ldots\ldots (1)$$

- Where,
- I is the input line current
  Ia is the armature current
  Ish is the shunt field current
- Equation (1) is the current equation.



notes.arpanadhikari7.com.np

Available on notes.arpanadhikari7.com.np

The voltage equations are written by using Kirchhoff's voltage law (KVL) for the field winding circuit.

$$V = I_{sh}R_{sh} \ldots\ldots\ldots (2)$$

For armature winding circuit the equation will be given as:

$$V = E + I_aR_a \ldots\ldots\ldots (3)$$

The power equation is given as:

- **Power input = mechanical power developed + losses in the armature + loss in the field.**

$$VI = P_m + I_a^2R_a + I_{sh}^2R_{sh} \ldots\ldots\ldots (4)$$

$$VI = P_m + I_a^2R_a + VI_{sh}$$

$$P_m = VI - VI_{sh} - I_a^2R_a = V(I - I_{sh}) - I_a^2R_a$$

$$P_m = VI_a - I_a^2R_a = (V - I_aR_a)I_a$$

$$P_m = EI_a \ldots\ldots\ldots (5)$$

notes.arpanadhikari7.com.np

Available on notes.arpanadhikari7.com.np

Multiplying equation (3) by Ia we get the following equations.

$$VI_a = EI_a + I_a^2R_a \ldots\ldots\ldots (6)$$

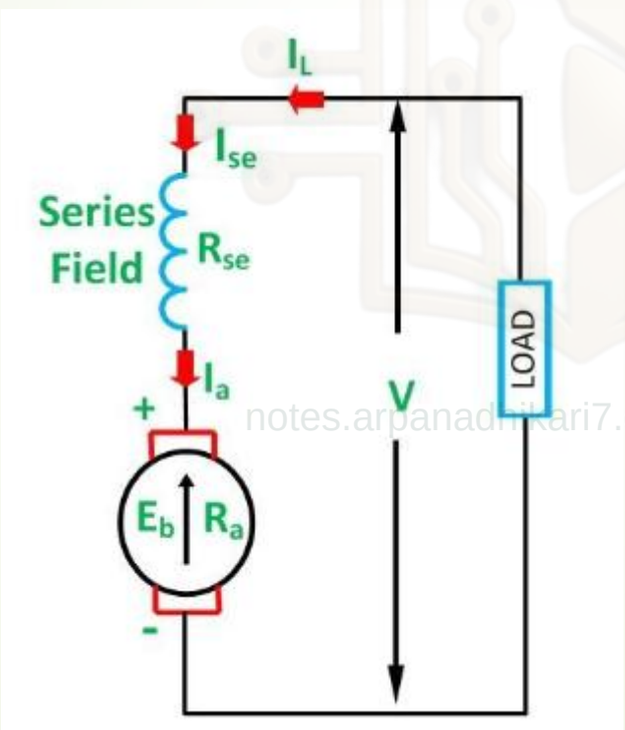
$$VI_a = P_m + I_a^2R_a \ldots\ldots\ldots (7)$$

Where,

VI_a is the electrical power supplied to the armature of the motor.

Series Wound Motor

- In the series motor, the field winding is connected in series with the armature winding.



notes.arpanadhikari7.com.np

Eb back emf

Available on notes.arpanadhikari7.com.np

By applying the KCL in the above figure:

$$I = I_{se} = I_a$$

Where,

- I_{se} is the series field current

The voltage equation can be obtained by applying KVL in the above figure.

$$V = E + I\,(R_a + R_{se}) \ldots\ldots\ldots (8)$$

The power equation is obtained by multiplying equation (8)

$$VI = EI + I^2\,(R_a + R_{se}) \ldots\ldots\ldots (9)$$

Power input = mechanical power developed + losses in the armature + losses in the field

$$VI = P_m + I^2R_a + I^2R_a \ldots\ldots\ldots (10)$$

notes.arpanadhikari7.com.np

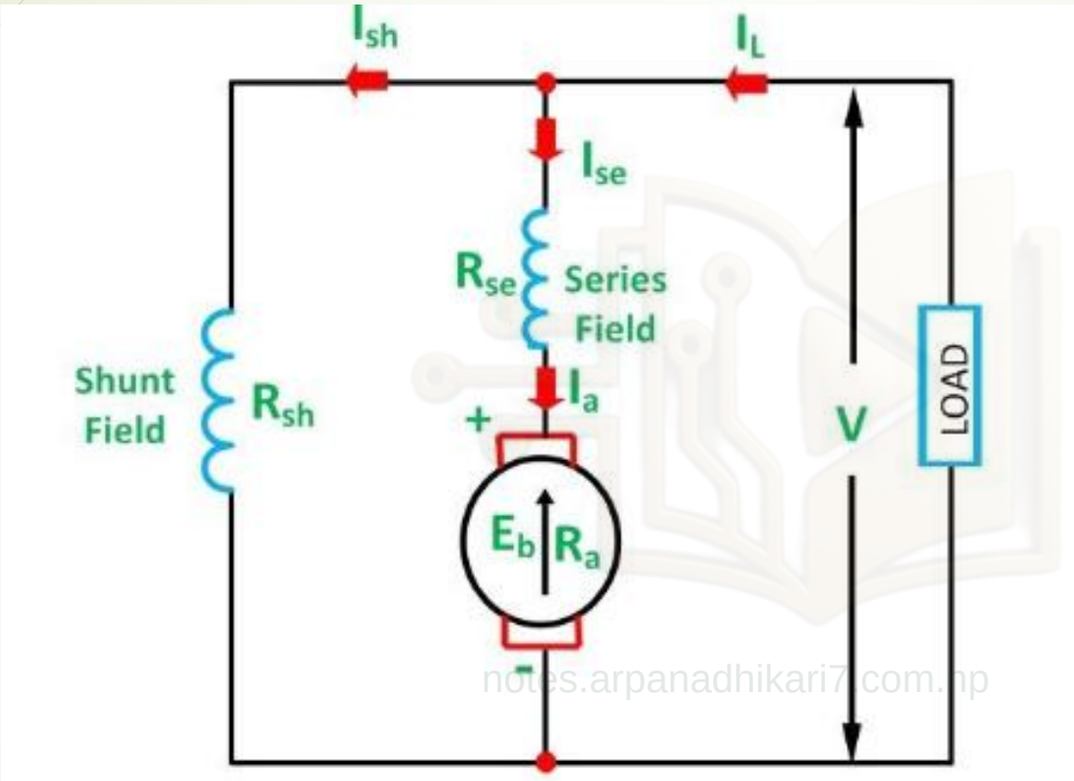
Comparing the equation (9) and (10), we will get the equation shown below

$$P_m = EI \ldots\ldots\ldots (11)$$

Available on notes.arpanadhikari7.com.np

Compound Wound Motor

A DC Motor having both shunt and series field windings is called a **Compound Motor**.



notes.arpanadhikari7.com.np

$$V - I_{SE}R_{SE} - I_aR_a - E_b = 0$$

Available on notes.arpanadhikari7.com.np

Speed Of A DC Motor (Theory)

- Back emf "e" of a DC motor is nothing but the induced emf in armature conductors due to rotation of the armature in magnetic field.
- Thus, the magnitude of E_b can be given by EMF equation of a DC generator.
- $$e = \phi P \frac{N}{60} X \frac{Z}{A} volts$$

$E \propto \phi . N$

$N \propto E$

$N \propto \frac{1}{\phi}$

E_b can also be given as,

$E_b = V - I_a R_a$

thus, from the above equations

$N = e\ 60\ A/(P\ \phi\ Z)$ but, for a DC motor A, P and Z are constants

Therefore, $N \propto K\ e/_{\phi}$ (where, K=constant)

notes.arpanadhikari7.com.np

This shows the **speed of a dc motor** is directly proportional to the back emf and inversely proportional to the flux per pole.

Available on notes.arpanadhikari7.com.np

Speed Control Of DC Motor

Flux Control Method:

- As the **speed of a dc motor** is inversely proportional to the flux per pole.
- Thus by decreasing the flux, speed can be increased and vice versa.
- To control the flux, a rheostat is added in series with the field winding
- In case of **shunt motors adding more resistance in series** with the field winding will **increase the speed as it decreases the flux.**

Armature Control Method

- **Speed of a dc motor** is **directly proportional to the back emf E_b** and $E_b = V - I_aR_a$.
- when supply voltage V and the armature resistance R_a are kept constant, then the speed is directly proportional to armature current I_a.
- Thus, if resistance is added in series with the armature, I_a decreases and, hence, the speed also decreases.
- Greater the resistance in series with the armature, greater the decrease in speed.

Available on notes.arpanadhikari7.com.np